Apache Kafka: Asynchronous Messaging for Seamless Systems

Apache Kafka is one of the most popular open source message brokers. Found in almost all microservices environments, it has become an important component of Big Data manipulation. This article gives a brief description of Apache Kafka, followed by a case study that demonstrates how it is used.

Have you ever wondered how e-commerce platforms are able to handle immense traffic without getting stuck? Ever thought about how OTT platforms are able to deliver content to millions of users, smoothly and simultaneously? The key lies in their distributed architecture.

A system designed around distributed architecture is made up of multiple functional components. These components are usually spread across several machines, which collaborate with each other by exchanging messages asynchronously over a network. Asynchronous messaging is what enables scalable, non-blocking communication among components, thereby allowing smooth functioning of the overall system.

Asynchronous messaging

The common features of asynchronous messaging are:

- The producers and consumers of the messages are not aware of each other. They join and leave the system without the knowledge of the others.
- A message broker acts as the intermediary between the producers and consumers.
- The producers associate each of the messages with a type, known as topic. A topic is just a simple string.
- It is possible that producers send messages on multiple